

Mobile Camera Intrinsic Calibration Report

LAB0-1: Camera Calibration Experiment

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Data status: An OpenCV calibration was completed using 18 real JPEG images from an iPhone 15 Pro rear camera. The checkerboard has 8×5 inner corners and 3 cm squares.

1 Objectives

This experiment uses Python and OpenCV to perform camera calibration for a mobile-camera image set captured with a planar checkerboard. Its objectives are:

1. to understand the relationships among the world, camera, image, and pixel coordinate systems and the pinhole camera model;
2. to distinguish intrinsic parameters from extrinsic parameters and understand the roles of focal length, principal point, and lens distortion coefficients;
3. to use checkerboard corner detection, subpixel refinement, and `cv2.calibrateCamera`;
4. to obtain the camera matrix, distortion coefficients, and per-view poses, and to evaluate the fit using reprojection error;
5. to assess how checkerboard coverage, camera pose, and corner quality affect the calibration result.

All numerical results in this report come from the recorded JPEG data and the official NPZ result file. No hardware or calibration value is inferred from a default configuration.

2 Principles

2.1 Coordinate Systems and the Pinhole Model

The world coordinate system describes points on the calibration board. The camera coordinate system is centered at the optical center. The image coordinate system represents continuous image-plane coordinates, whereas the pixel coordinate system uses discrete image coordinates referenced to the upper-left image corner. Let $\mathbf{X}_w = (X_w, Y_w, Z_w)^\top$ be a point in world coordinates and $\mathbf{X}_c = (X_c, Y_c, Z_c)^\top$ the same point in camera coordinates. The extrinsic parameters are a rotation matrix $\mathbf{R}$ and a translation vector $\mathbf{t}$:

$$\mathbf{X}_c = \mathbf{R}\mathbf{X}_w + \mathbf{t}. \quad (1)$$

Under the pinhole model, let $x = X_c/Z_c$ and $y = Y_c/Z_c$ be normalized image coordinates. The projection to pixel coordinates is

$$s \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \mathbf{K} [\mathbf{R} \quad \mathbf{t}] \begin{bmatrix} X_w \\ Y_w \\ Z_w \\ 1 \end{bmatrix}, \quad s \neq 0, \quad (2)$$

where (u, v) are pixel coordinates and $\mathbf{K}$ is the camera matrix:

$$\mathbf{K} = \begin{bmatrix} f_x & \gamma & c_x \\ 0 & f_y & c_y \\ 0 & 0 & 1 \end{bmatrix}. \quad (3)$$

Here f_x and f_y are focal lengths in pixels, (c_x, c_y) is the principal point, and γ is the skew parameter. For ordinary mobile-camera images, γ is commonly close to zero.

2.2 Lens Distortion Model

Real lenses make the observed image deviate from ideal pinhole projection. Let $r^2 = x^2 + y^2$. A common radial distortion model is

$$\begin{aligned} x_r &= x (1 + k_1 r^2 + k_2 r^4 + k_3 r^6), \\ y_r &= y (1 + k_1 r^2 + k_2 r^4 + k_3 r^6). \end{aligned} \quad (4)$$

The radial distortion coefficients are k_1, k_2, k_3 . Tangential distortion can be written as

$$\begin{aligned} x_d &= x_r + 2p_1 xy + p_2 (r^2 + 2x^2), \\ y_d &= y_r + p_1 (r^2 + 2y^2) + 2p_2 xy. \end{aligned} \quad (5)$$

The final pixel coordinates are

$$u = f_x x_d + \gamma y_d + c_x, \quad v = f_y y_d + c_y. \quad (6)$$

OpenCV commonly returns five coefficients in the order $(k_1, k_2, p_1, p_2, k_3)$. If an interface returns a different length, the report must preserve the returned order and length.

2.3 Planar Calibration and the Zhang Method

Set the checkerboard plane to $Z_w = 0$ and define board coordinates using the square size. The mapping from a point on the plane to its image is a homography:

$$s \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \mathbf{H} \begin{bmatrix} X_w \\ Y_w \\ 1 \end{bmatrix}, \quad \mathbf{H} = \mathbf{K} [\mathbf{r}_1 \quad \mathbf{r}_2 \quad \mathbf{t}], \quad (7)$$

where $\mathbf{r}_1$ and $\mathbf{r}_2$ are the first two columns of the rotation matrix. Multiple views of the same checkerboard provide homography constraints. The orthogonality and unit-length constraints of the rotation columns jointly estimate the intrinsic parameters and then the pose of each view. A nonlinear optimization finally refines the intrinsic parameters, distortion coefficients, and extrinsic parameters by minimizing the discrepancy between observed and projected corners.

In the program, the checkerboard object points lie on $Z_w = 0$, and the detected image points form the corresponding two-dimensional observations. The function `cv2.calibrateCamera` performs the calibration from these correspondences. The checkerboard configuration and square size must match the physical board; otherwise the translation-vector scale is not physically meaningful.

2.4 Reprojection Error

Let $\mathbf{q}_{j,i}$ be the observed pixel for corner i in image j , and let $\hat{\mathbf{q}}_{j,i}$ be the pixel obtained by projecting the estimated three-dimensional point with `cv2.projectPoints`. The pointwise reprojection error is

$$e_{j,i} = \|\hat{\mathbf{q}}_{j,i} - \mathbf{q}_{j,i}\|_2. \quad (8)$$

If M is the total number of valid corners, the mean pointwise reprojection error and global RMS are

$$\bar{e} = \frac{1}{M} \sum_{j,i} e_{j,i}, \quad e_{\text{RMS}} = \sqrt{\frac{1}{M} \sum_{j,i} e_{j,i}^2}. \quad (9)$$

For image j , the per-image RMS and its arithmetic mean are

$$e_{j,\text{RMS}} = \sqrt{\frac{1}{M_j} \sum_i e_{j,i}^2}, \quad \bar{e}_{\text{view}} = \frac{1}{n} \sum_{j=1}^n e_{j,\text{RMS}}, \quad (10)$$

where M_j is the number of corners in image j and n is the number of valid images. The NPZ fields `rms_error`, `mean_reprojection_error`, and `mean_per_image_rms` record these three distinct quantities.

Table 1: Experimental environment and acquisition record

Category	Item	Recorded value
Operating system	System and architecture	macOS 26.6.2, Darwin 25.6.0, arm64
Software	Python	3.13.12
Software	OpenCV	4.10.0 (cv2)
Software	Main APIs	<code>findChessboardCorners</code> , <code>cornerSubPix</code>
Software	Main APIs	<code>calibrateCamera</code> , <code>projectPoints</code>
Hardware	Phone model	Apple iPhone 15 Pro (JPEG EXIF)
Hardware	Camera mode	Rear camera; focal length not recorded
Image	Image resolution	5712×4284
Board	Inner-corner configuration	8×5
Board	Physical square size	3 cm
Data	Input images	18 JPEG files (<code>task2/images_jpg/</code>)
Data	Valid/failed images	18 / 0

3 Experimental Environment

The software environment was verified directly. The Python interpreter was `/Users/jiaqiaosu/miniconda3/bin/python`; it imports `cv2`. The JPEG EXIF records an Apple iPhone 15 Pro, and the data were captured with its rear camera. The specific rear-lens focal length was not recorded and is not inferred here.

The software versions, JPEG device information, image size, checkerboard configuration, 3 cm square size, and image counts were verified from local commands, EXIF, acquisition records, and `camera_params.npz`. The program was run with `square_size=3.0`, so the translation vectors are measured in cm. The specific rear-lens focal length was not recorded.

4 Data Acquisition

4.1 Checkerboard and Coordinate Convention

The calibration target is a planar black-and-white checkerboard. The option `--checkerboard` specifies the number of checkerboard inner-corner columns and rows, while `--square-size` specifies the physical side length of one square. The present run used 8×5 inner corners and a 3 cm square size, with `square_size=3.0`. The upper-left inner corner was set to $(0, 0, 0)$; the two board directions define X_w and Y_w , and all board points have $Z_w = 0$. This convention is consistent with the planar model in Equation 7.

4.2 Acquisition Conditions

The input set contains 18 JPEG files under `task2/images_jpg/`, each with resolution 5712×4284 . JPEG EXIF records an Apple iPhone 15 Pro, and the acquisition used its rear camera. The specific rear-lens focal length was not recorded and is not inferred. The original HEIC files under `task2/images/` were not used as direct OpenCV input because the current `cv2.imread` cannot read them.

The board should be captured at different distances, translations, and tilt angles so that it covers the image center, edges, and different depths. A set consisting only of nearly frontal views with repeated poses is insufficient. The board should be complete and sharp, without severe reflection, motion blur, occlusion, or overexposure. A separate pose-coverage log remains to be organized from the original acquisition.

The program reads `task2/images_jpg/` and preserves the original filenames, allowing the NPZ `image_paths` field to be matched to each detection result.

The calibration values come from the actual run on the 18 JPEG files. The rear-camera mode, checkerboard configuration, and 3 cm square size are fixed acquisition facts; only the exact rear-lens focal length and the detailed pose-coverage log remain undocumented.

Table 2: Mobile-camera acquisition record

Item	Recorded content	Status
Phone	Manufacturer and model	Apple iPhone 15 Pro (JPEG EXIF)
Camera	Front/rear camera and lens mode	Rear camera; focal length not recorded
Image resolution	Width $\times$ height (pixels)	5712 $\times$ 4284
Image count	Input JPEG files	18
Checkerboard	Inner corners per row/column	8 $\times$ 5
Square size	Physical side length	3 cm; <code>square_size=3.0</code>
Pose coverage	Position, distance, and tilt	Acquisition log to be organized

5 Experimental Procedure

1. Prepare the planar checkerboard, measure the physical square size, and record the number of inner corners in each direction.
2. Capture clear images at different distances, translations, and tilt angles. Record the phone model, rear-camera use, image resolution, and acquisition settings.
3. Read each image and check whether it can be loaded. A read failure is recorded for that file and does not stop processing of the remaining images.
4. Convert a color image to grayscale with `cv2.cvtColor`, then detect checkerboard inner corners with `cv2.findChessboardCorners`. Record the two-dimensional corners for successful images and the filename and reason for failures.
5. Refine detected corners to subpixel accuracy with `cv2.cornerSubPix`.
6. Generate planar three-dimensional object points from the checkerboard configuration and square size, and pair them with the two-dimensional image points. The geometry follows Equation 7.
7. Check the number of valid images and the consistency of image sizes. Mixed resolutions and fewer valid views than `--min-images` must be rejected.
8. Call `cv2.calibrateCamera` to estimate the camera matrix, distortion coefficients, and the rotation and translation vectors for each valid view. The camera matrix is defined in Equation 3; the distortion model is given in Equations 4 and 5.
9. Reproject the three-dimensional points with `cv2.projectPoints`, compute pointwise errors using Equation 8, and summarize the three error definitions in Section 7.
10. Save the camera matrix, distortion coefficients, poses, error statistics, image counts, and detection status to the NPZ result. If requested, use `cv2.undistort` to save corrected images.
11. Use the recorded result and the criteria in Section 8 to assess parameter consistency and data quality.

The complete control flow implemented by the OpenCV program is summarized in Figure 1.

6 Program Design

6.1 Input, Configuration, and Output

The Python entry point uses `argparse` and reads JPEG files from `task2/images_jpg/`.

The options `--checkerboard` and `--square-size` specify the checkerboard inner-corner configuration and square size. The options `--images` and `--output-dir` specify the input pattern and output directory. The option `--min-images` sets the minimum number of valid views.

The present run used 8 $\times$ 5 inner corners, `square_size=3.0` cm, and `--min-images 3`. Each image was read, converted to grayscale, searched for corners, and refined to subpixel accuracy. Successful files print `[OK]`; read failures, corner failures, and resolution mismatches print `[FAILED]` with a reason and are counted. All 18 JPEG files succeeded in this run. HEIC files under `task2/images/` are not directly readable by the current `cv2.imread`.

6.2 Core Computation

Three-dimensional board points are generated from the checkerboard geometry, while two-dimensional points are obtained from the refined corners. Calling `cv2.calibrateCamera` produces the camera matrix, distortion coefficients, and a rotation and translation vector for each valid view. The call receives `objectPoints`,

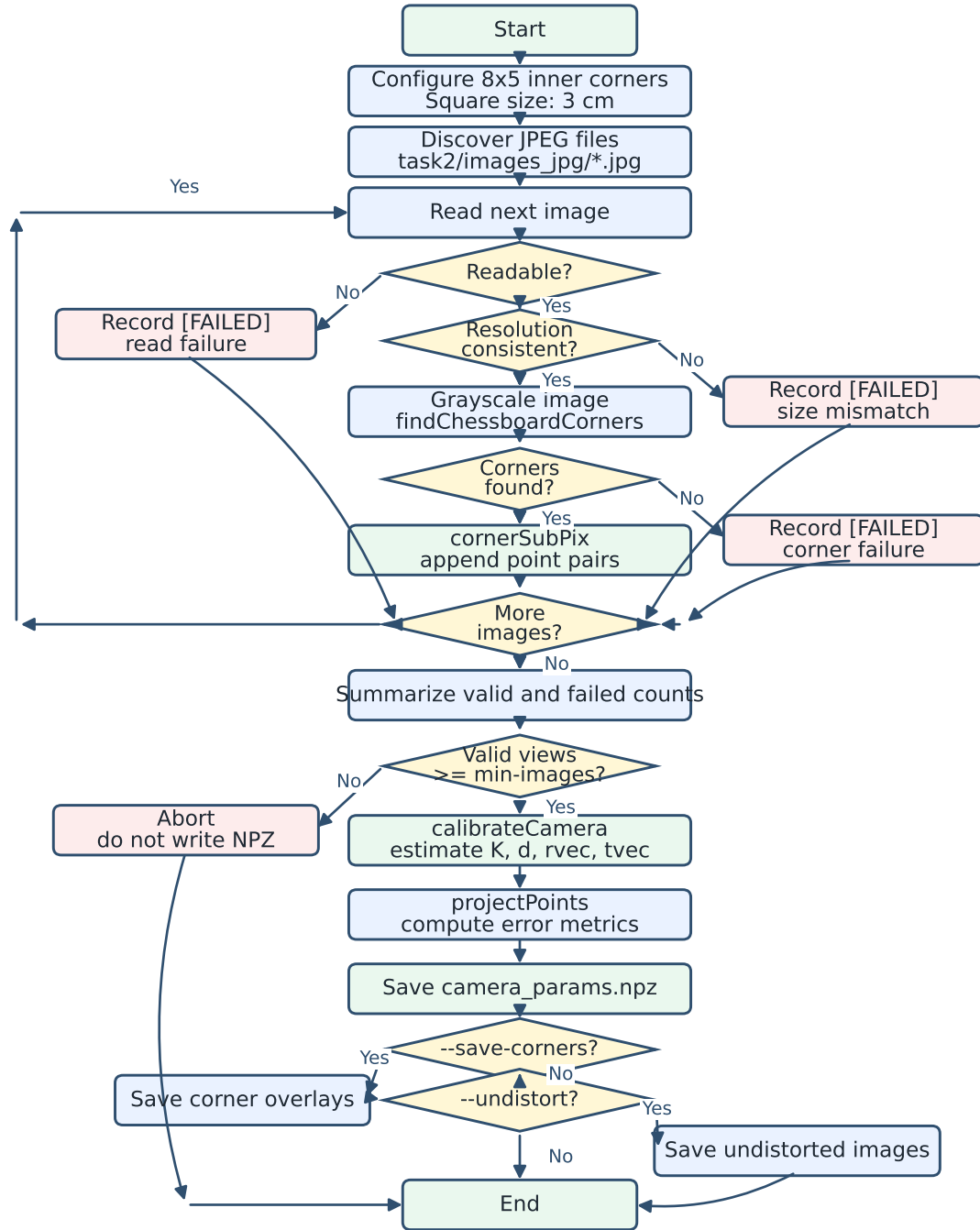


Figure 1: OpenCV camera-calibration workflow implemented in this experiment.

`imagePoints`, and `imageSize`; the projection relation is given in Equation 2.

The program uses `cv2.projectPoints` to reproject every board point and compare it with the detected pixel. It stores OpenCV RMS in `rms_error`, the per-image RMS array in `per_image_reprojection_error`, the global pointwise mean in `mean_reprojection_error`, and the arithmetic mean of per-image RMS values in `mean_per_image_rms`. Their definitions are given in Equations 9 and 10.

The official result is written to `task2/output/camera_params.npz`. The file records the image size, checkerboard configuration, poses, candidate/valid/failed paths, failure reasons, and all three error statistics. The 3.0 square-size value is in cm, so the translation vectors are in cm. Hardware mode and physical board measurements are acquisition facts, not values inferred by the program.

7 Experimental Results

The results below were read directly from `task2/output/camera_params.npz` for the 18 real JPEG files in `task2/images_jpg/*.jpg`. The run used an 8×5 checkerboard, a 3.0 cm square size, and `--min-images 3`. These values are recorded outputs, not estimates or placeholders. The image order used in the per-view figures is the order of `valid_image_paths` in the NPZ file.

7.1 Recovered calibration quantities

The recovered camera matrix is

$$\mathbf{K} = \begin{bmatrix} 4082.15290893 & 0 & 2883.66417823 \\ 0 & 4083.58657686 & 2107.20293734 \\ 0 & 0 & 1 \end{bmatrix}. \quad (11)$$

The five-coefficient OpenCV distortion vector is

$$\mathbf{d} = [0.20245123; -0.29420837; -0.00326679; 0.00308671; -1.04554325], \quad (12)$$

where $\mathbf{d}$ follows the order $(k_1, k_2, p_1, p_2, k_3)$.

Table 3: Mobile-camera calibration results read from the official NPZ file

Category	Metric	Recorded value
Data	Input pattern	<code>task2/images_jpg/*.jpg</code>
Data	Candidate images	18
Data	Valid calibration images	18
Data	Failed images	0 (read, corner, and size failures)
Data	Image size (width $\times$ height)	5712×4284
Configuration	Checkerboard inner corners	8×5
Configuration	<code>square_size</code>	3.0 cm
Intrinsics	Camera matrix $\mathbf{K}$	See Equation 11
Intrinsics	f_x	4082.15290893
Intrinsics	f_y	4083.58657686
Intrinsics	c_x	2883.66417823
Intrinsics	c_y	2107.20293734
Distortion	$(k_1, k_2, p_1, p_2, k_3)$	See Equation 12
Error	OpenCV RMS (pixels)	2.06424401
Error	Mean pointwise reprojection error (pixels)	1.65696931
Error	Mean per-image RMS (pixels)	1.90750642
Extrinsics	Rotation/translation shape	$18 \times 3 \times 1$; translations in cm

OpenCV RMS is the global root-mean-square error returned by `cv2.calibrateCamera`. The mean pointwise reprojection error is `mean_reprojection_error`, averaged over all $18 \times 40 = 720$ valid corners. The mean per-image RMS is `mean_per_image_rms`, the arithmetic mean of 18 per-image RMS values. These definitions are distinct. The per-image array is stored in `per_image_reprojection_error`; poses are stored in `rotation_vectors` and `translation_vectors`.

7.2 Per-view and geometric diagnostics

The global intrinsic matrix and distortion vector are each one joint estimate from all 18 views. They are not 18 independent estimates. Therefore, the additional plots below show quantities that genuinely vary by view: reprojection residuals, checkerboard pose, detected-corner coverage, and the response of the single fitted distortion model. This distinction prevents visual fluctuation plots from incorrectly treating one global $\mathbf{K}$ or $\mathbf{d}$ as if it had been refitted independently to every image.

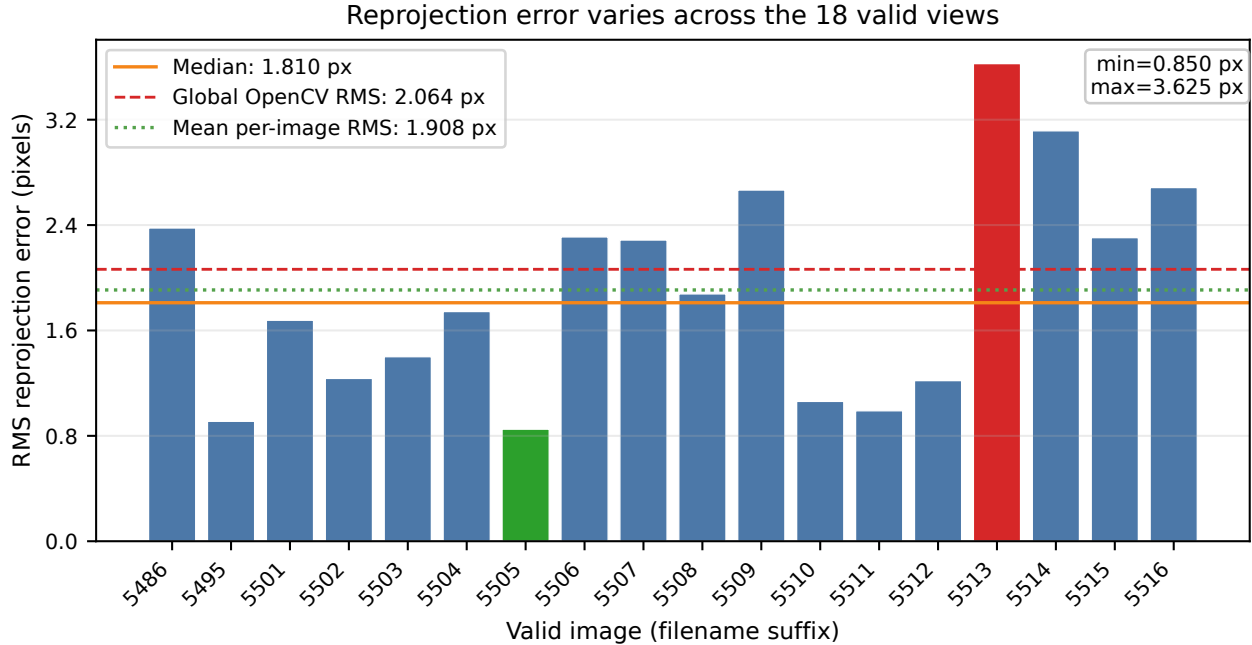


Figure 2: Per-image RMS reprojection error for the 18 valid views. The bars are calculated from the 40 detected corners in each image using `cv2.projectPoints`. The horizontal lines show the median, the arithmetic mean of per-image RMS values, and the global OpenCV RMS. These bars are per-view residual diagnostics, not 18 separate intrinsic-parameter fits.

The per-image RMS values range from 0.850436 px (IMG_5505) to 3.625463 px (IMG_5513), with a median of 1.810399 px and an interquartile range of 1.223223–2.360739 px. Eight of the 18 views exceed 2 px, and four exceed 2.5 px. These views remain valid detections, but their larger residuals make them useful candidates for visual inspection or for a future refined acquisition. The plot also shows why the arithmetic mean of the per-image RMS values (1.907506 px) is not identical to the global OpenCV RMS (2.064244 px).

The detected corners occupy normalized coordinates approximately from (0.191, 0.182) to (0.814, 0.792), so the union of the observed board corners covers about 62.3% of the image width and 61.1% of the image height. This is useful central coverage, but it does not reach the outer image border. The fitted distortion coefficients are consequently less directly constrained at the extreme corners than in the central region, which is an important limitation when interpreting the distortion-response plot below.

The checkerboard-origin depth t_z varies from 24.705 to 90.828 cm, and the board-origin distance $\|\mathbf{t}\|$ varies from 27.559 to 91.815 cm. The lateral components span -17.256 to 7.327 cm in x and -12.834 to 10.166 cm in y . Thus the dataset contains both distance and lateral-position changes rather than a repeated single pose.

The Rodrigues-vector norm ranges from 2.688° to 173.651° , while the board-normal tilt ranges from 2.683° to 62.936° . The two angles answer different questions: the norm describes the axis-angle representation of the full board-to-camera rotation, whereas the board-normal tilt isolates the departure of the board plane from a fronto-parallel view. The large norm in IMG_5516 should therefore be interpreted together with the other components and the original image, not automatically labeled as a calibration failure.

The distortion plot gives an interpretable view of the coefficient signs and magnitudes beyond listing five numbers. At the largest normalized radius corresponding to the image corners, the radial scale factor predicted by the polynomial is approximately 0.476, while the tangential displacement remains on the order of 10^{-2} normalized coordinate units and depends on angular direction. This strong high-radius response should be treated cautiously because the observed-corner coverage does not reach the sensor border. It is a model extrapolation that should be checked against additional edge-of-frame calibration views before using the parameters for

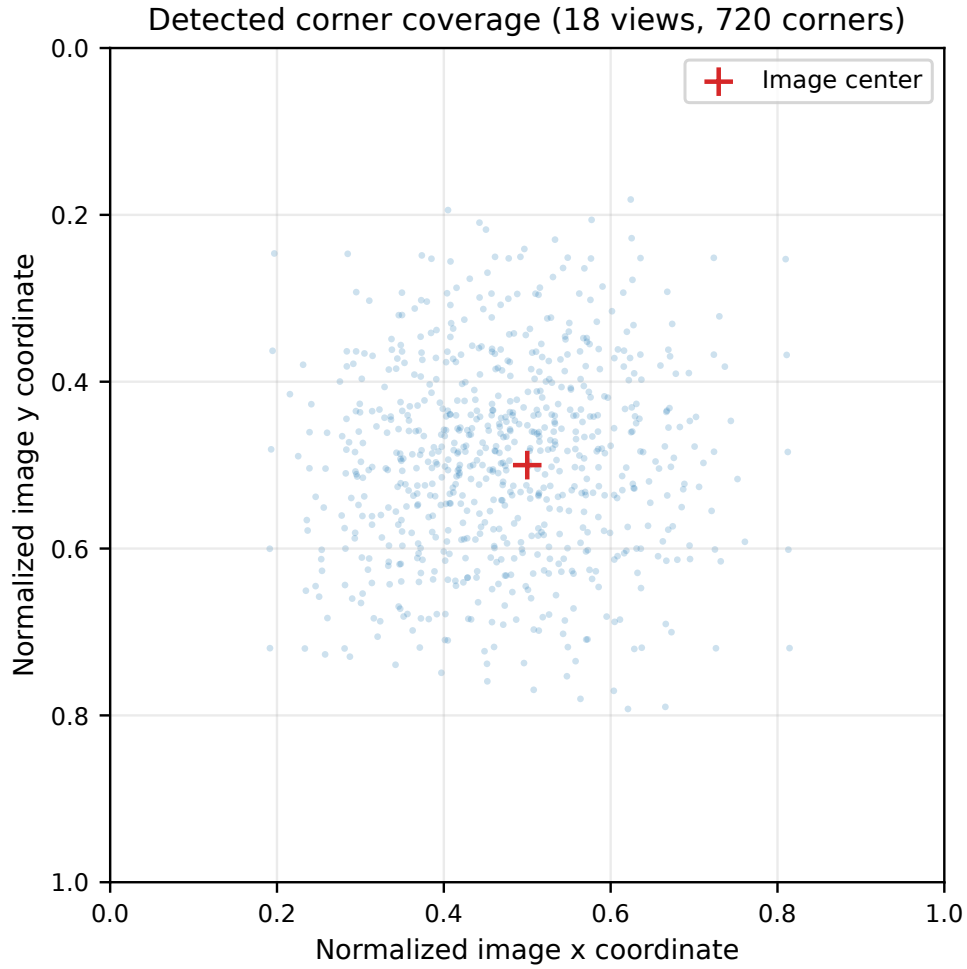


Figure 3: Coverage of the 720 refined checkerboard corners in normalized image coordinates. The points were regenerated with the same `findChessboardCorners` flags and `cornerSubPix` criteria used by the calibration program. The plot summarizes where observations were actually made; it is not a claim that the entire sensor area was sampled uniformly.

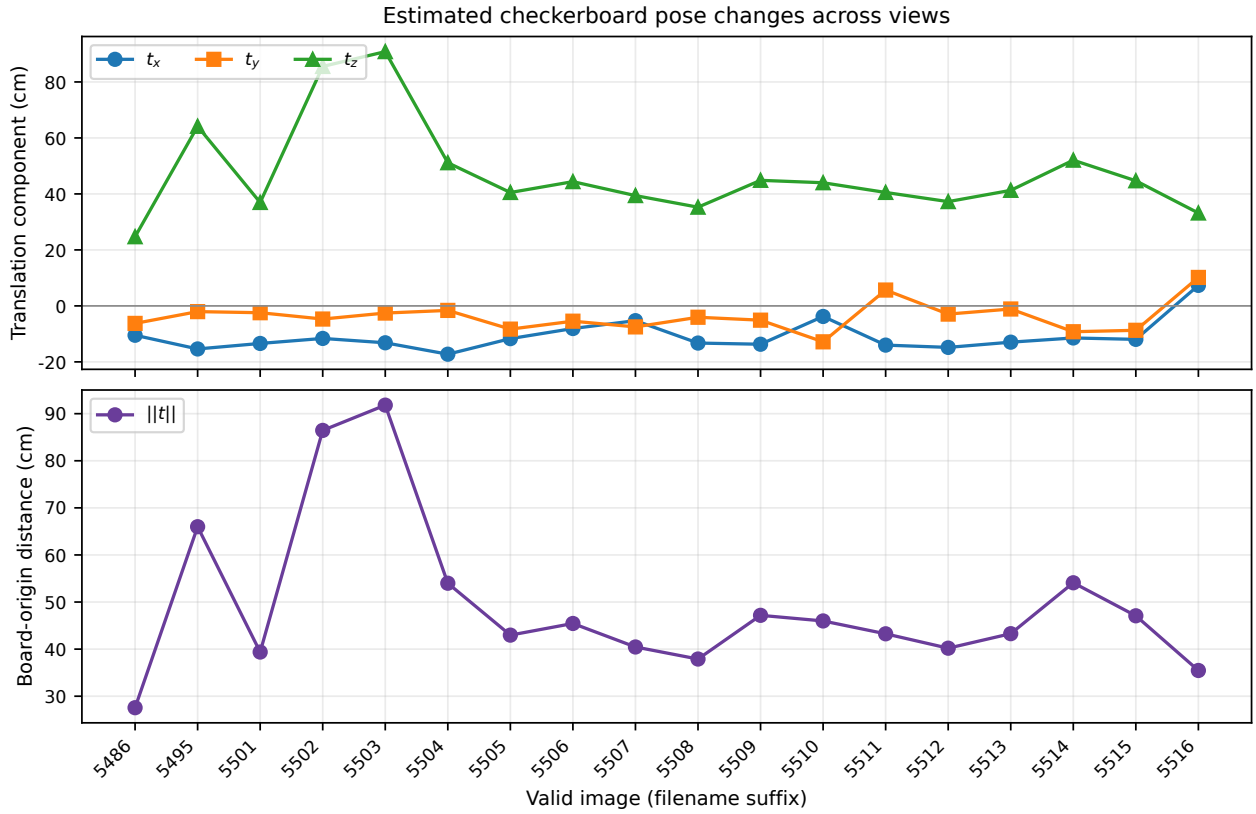


Figure 4: Translation-vector variation across views. OpenCV uses $\mathbf{t}$ to map the checkerboard origin from board coordinates to camera coordinates; because the object-point square size is 3 cm, the plotted components and $\|\mathbf{t}\|$ are in centimeters. The t_z curve is the depth of the board origin, while the lower panel gives its Euclidean distance from the camera origin.

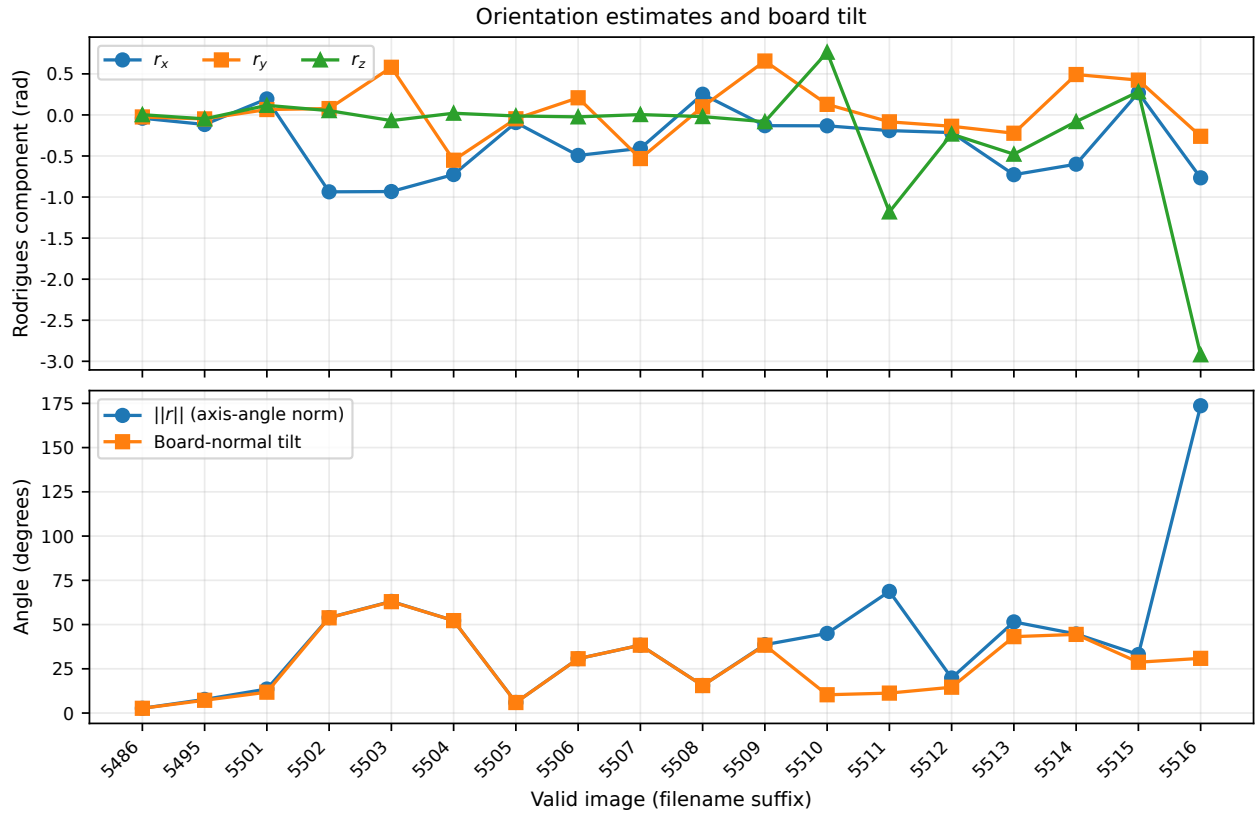


Figure 5: Rotation-vector and board-tilt diagnostics. The upper panel reports the three OpenCV Rodrigues-vector components in radians. The lower panel reports the Rodrigues-vector norm in degrees and the angle between the transformed board normal and the camera optical axis. The latter is a direct board-tilt diagnostic and should not be confused with camera roll.

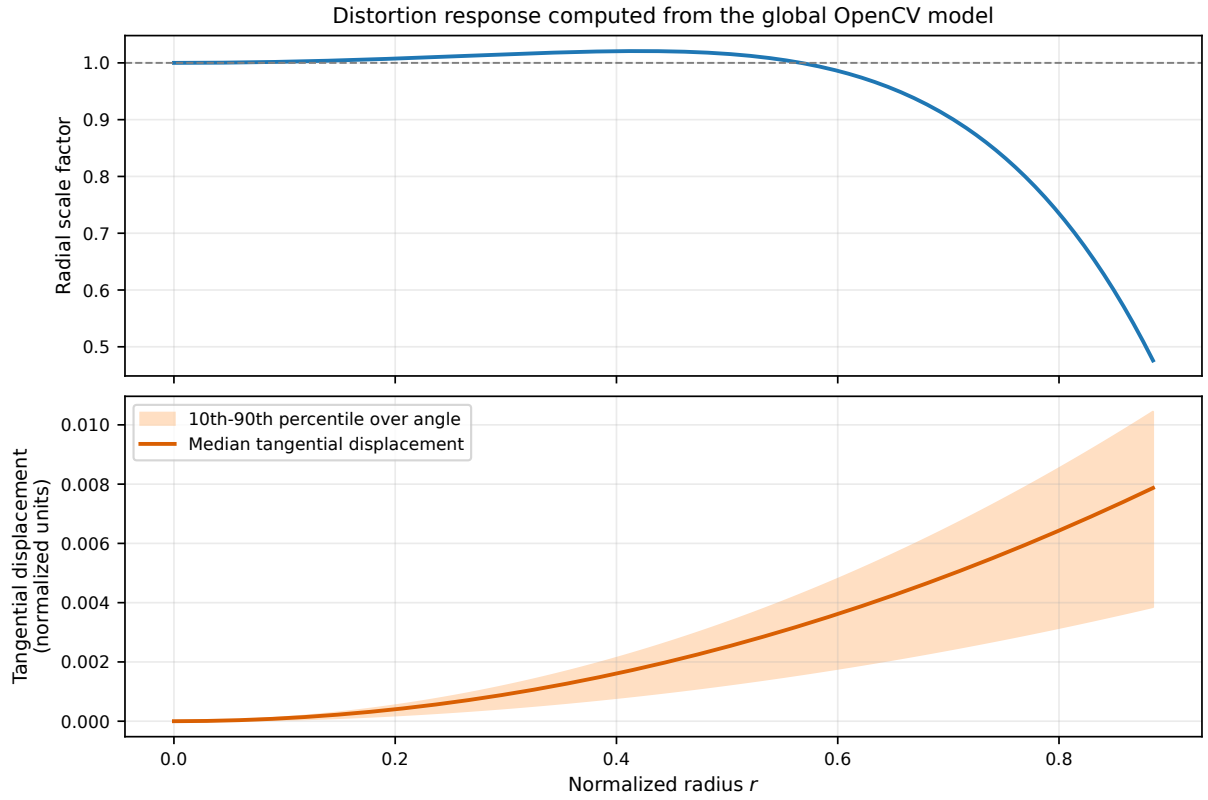


Figure 6: Response of the fitted distortion model over normalized radius. The upper curve is the radial scale factor $1 + k_1 r^2 + k_2 r^4 + k_3 r^6$. The lower curve is the tangential displacement induced by (p_1, p_2) , summarized over angular direction. Both panels are deterministic calculations from the one global coefficient vector in Equation 12; they are not per-image measurements.

precision measurements.

Table 4: Additional measured ranges used in the diagnostic analysis

Quantity	Measured summary	Meaning
Per-image RMS	0.850–3.625 px; median 1.810 px	View-dependent residual quality
Board depth t_z	24.705–90.828 cm	Distance variation along optical axis
Board distance $\ \mathbf{t}\ $	27.559–91.815 cm	Distance from camera origin
Board-normal tilt	2.683–62.936°	Plane-orientation variation
Corner union	62.3% width, 61.1% height	Observed image-area span

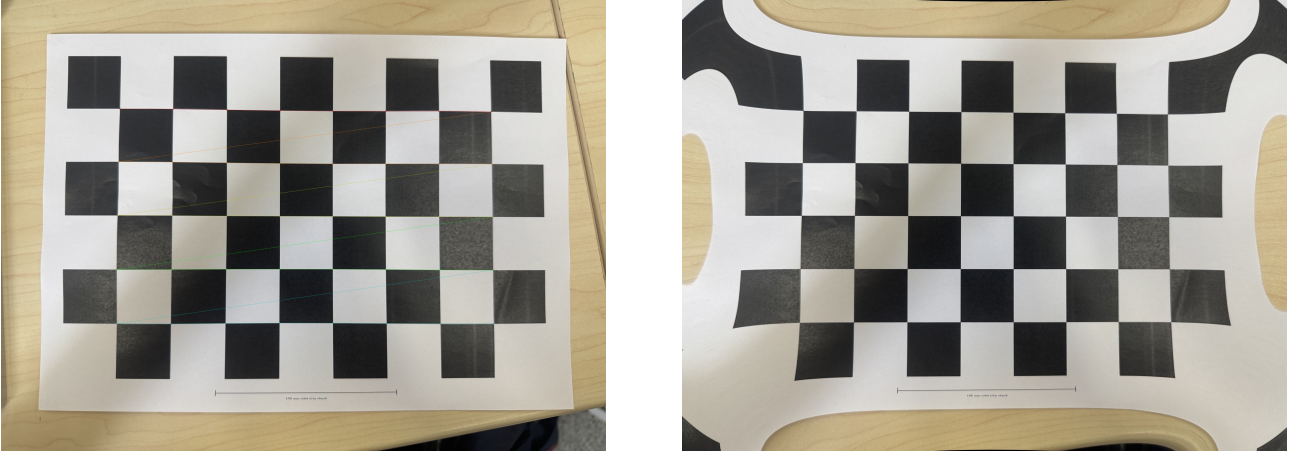


Figure 7: Representative results for `IMG_5486.jpg`: the left image shows the detected corners, and the right image is the undistorted image produced with the same calibration. The figures come from the temporary visualization run; numerical results remain those in `task2/output/camera_params.npz`.

Figure 7 shows one of the 18 valid views rather than repeating all outputs. The corner overlay checks that the expected 8×5 inner-corner pattern was found, while the corrected image provides a qualitative check on the fitted distortion model. A single visual example cannot validate all views, so the quantitative per-view plots and the coverage plot are retained as complementary checks.

8 Analysis

This analysis is based on the 18 real JPEG images, the official result file, and the corresponding OpenCV run. Conclusions are limited to the current 5712×4284 image-coordinate setup, 8×5 checkerboard inner corners, and 3 cm square size. A pixel reprojection error is not treated as a direct guarantee of physical measurement accuracy.

8.1 Scope of the plotted variation

The camera matrix $\mathbf{K}$ and distortion vector $\mathbf{d}$ are fitted jointly from all valid views, so this experiment produces one global estimate of f_x, f_y, c_x, c_y and $(k_1, k_2, p_1, p_2, k_3)$. It would be methodologically incorrect to draw 18 apparent intrinsic-parameter trajectories by copying the same global values or by fitting a single planar view independently without reporting a separate uncertainty protocol. The additional figures therefore use quantities that really have one value per image: reprojection RMS, rotation and translation vectors, detected-corner locations, and the response of the fitted distortion model over normalized radius. This gives more diagnostic information without fabricating parameter fluctuations.

8.2 Intrinsic-parameter consistency

The estimated f_x, f_y, c_x , and c_y are finite and positive. From the measured focal lengths,

$$\rho_f = \frac{|f_x - f_y|}{(f_x + f_y)/2} = 0.0003511422 \quad (0.035114\%).$$

The two focal-length estimates are therefore very close, which is compatible with nearly equal horizontal and vertical pixel scales in this image set. It does not identify the exact physical focal length of the iPhone lens, and it does not rule out digital cropping, resizing, or other image-processing operations.

The image center is

$$(W/2, H/2) = (2856, 2142),$$

whereas the principal-point offset is

$$(c_x - W/2, c_y - H/2) = (27.66418, -34.79706) \text{ px}.$$

Its Euclidean length is 44.45382 px, approximately 0.7783% of the image width and 1.0377% of its height. The principal point is thus near, but not identical to, the image center. This is a reasonable result for a calibrated image coordinate system, but the offset should not be interpreted as a hardware defect without controlling the phone’s crop and image pipeline.

8.3 Per-view reprojection error

The global OpenCV RMS is 2.06424401 px, the global mean pointwise distance is 1.65696931 px, and the arithmetic mean of the 18 per-image RMS values is 1.90750642 px. The per-image distribution in Figure 2 has minimum 0.850436 px, first quartile 1.223223 px, median 1.810399 px, third quartile 2.360739 px, maximum 3.625463 px, and sample standard deviation 0.811881 px. The largest residuals occur in IMG_5513 (3.625463 px), IMG_5514 (3.115401 px), IMG_5516 (2.684612 px), and IMG_5509 (2.665750 px).

These values indicate that corner localization quality is not identical across views, even though all images passed the program’s validity checks. A higher residual can result from blur, glare, partial perspective compression, printing or board flatness, or local corner ambiguity; the residual alone cannot identify which cause occurred. The per-view errors have an exploratory Pearson correlation of approximately $r = 0.38$ with board-normal tilt and $r = -0.32$ with board-origin distance. With only 18 observations, these weak correlations are not evidence of a causal trend, but they suggest that strongly tilted views deserve visual inspection. Removing such a view without rerunning calibration and documenting the effect would also be premature.

At the current resolution, the OpenCV RMS is approximately 0.03614% of the image width and 0.04818% of its height. Thus the error is small relative to the full image dimensions, but a high-resolution image makes a fixed pixel error appear smaller in relative terms. The RMS is an internal fit residual, not an independent validation measurement, and comparisons across resolutions should account for image scale and acquisition conditions.

8.4 Pose diversity and calibration geometry

The translation plot in Figure 4 shows meaningful geometric variation. The checkerboard-origin depth t_z ranges from 24.705 to 90.828 cm, while $\|\mathbf{t}\|$ ranges from 27.559 to 91.815 cm. The lateral components span 24.583 cm in x and 23.001 cm in y . The board was therefore not observed at one fixed distance and one fixed lateral location.

The orientation plot in Figure 5 gives a complementary view. The Rodrigues-vector norm ranges from 2.688° to 173.651° , and the board-normal tilt ranges from 2.683° to 62.936° , with a median tilt of 29.703° . The Rodrigues vector is an axis-angle representation of the board-to-camera rotation; its component values and norm are not three independent physical tilt angles. The board-normal tilt is easier to interpret as the angle between the board plane normal and the camera optical axis. The near- 180° norm in IMG_5516 may reflect the rotation representation and the board’s in-plane orientation, so it should be checked against the image rather than automatically treated as a bad calibration pose.

8.5 Corner coverage and distortion extrapolation

All 18 candidate JPEG files passed image reading, 8×5 corner detection, and the resolution check. Every valid image contributes 40 refined corners, giving 720 observations. Figure 3 shows that the union of those corners extends from approximately $x = 0.191$ to 0.814 and $y = 0.182$ to 0.792 in normalized image coordinates. In physical pixels, the board observations therefore occupy a central region rather than the full 5712×4284 sensor frame.

This coverage is sufficient for a working calibration run, but it limits the evidence available for the outermost distortion behavior. Figure 6 is computed from the global coefficient vector, not from measured corner

displacements at every radius. At the image-corner radius in the current camera model, the radial scale factor is approximately 0.476, while the tangential displacement is approximately 0.004–0.011 normalized coordinate units over viewing angle. The radial polynomial’s strong high-radius response may be real, but the present dataset does not densely observe that region. Additional views placing the checkerboard near the left, right, top, bottom, and four corner regions would be the most valuable next experiment.

8.6 Distortion interpretation

In OpenCV order, the distortion vector is

$$(k_1, k_2, p_1, p_2, k_3) = (0.20245123, -0.29420837, -0.00326679, 0.00308671, -1.04554325).$$

The nonzero p_1 and p_2 terms indicate that the fitted model includes tangential distortion, while the alternating signs of the radial terms make the radial response nonlinear rather than a simple monotonic correction. The numerical size of a distortion coefficient is scale-dependent because it is applied to normalized coordinates; it should not be compared with coefficients obtained under a different resolution or coordinate normalization without checking the model definition.

The undistorted example in Figure 7 is a qualitative check only. A stronger validation would hold out several images, estimate parameters on the remaining views, and compare reprojection errors on the held-out corners. That uncertainty or cross-validation analysis was not part of the current run, so the report does not claim it.

8.7 Limitations and practical conclusion

The experiment demonstrates a reproducible OpenCV calibration for the supplied iPhone 15 Pro rear-camera images, with translations expressed in centimeters because `square_size=3.0` was supplied in centimeters. The result is tied to the selected rear-camera mode, resolution, crop, focus, and computational photography pipeline. The NPZ file does not contain a covariance estimate for the global parameters, and the current acquisition does not document a per-image pose label independent of the estimated extrinsics. Consequently, the recovered $\mathbf{K}$ and $\mathbf{d}$ are suitable as calibrated parameters for this image-coordinate setup, but they should be re-estimated if the phone changes lens mode, resolution, crop, or focus. The most direct improvement is to add more edge and corner views, keep the board sharp and flat, and compare the new per-view residual distribution with Figure 2.

9 Conclusion

This experiment completed OpenCV camera calibration using 18 real JPEG images from the rear camera of an iPhone 15 Pro. The calibration used an 8×5 checkerboard-inner-corner configuration and a 3 cm square size. The program produced the camera matrix, five distortion coefficients, 18 rotation vectors, and 18 translation vectors, and recorded OpenCV RMS, mean pointwise reprojection error, and mean per-image RMS. All 18 candidate images were valid; no image-read, corner-detection, or resolution failures occurred.

The run used `square_size=3.0`, so translation vectors are expressed in cm, while the camera matrix and pixel-coordinate distortion results describe the present image model. JPEG EXIF and the acquisition record identify the Apple iPhone 15 Pro rear camera. The exact rear-lens focal length was not recorded and is not inferred here. A detailed pose-coverage log should still be added before submission, but it does not affect the numerical readback of the official NPZ result.